

CYBERSECURITY PROTECTS DIGITAL INFRASTRUCTURE

Target Audience: Beginners

Defending computers, servers, and data from malicious attacks in our hyper-connected world.

THE CIA TRIAD

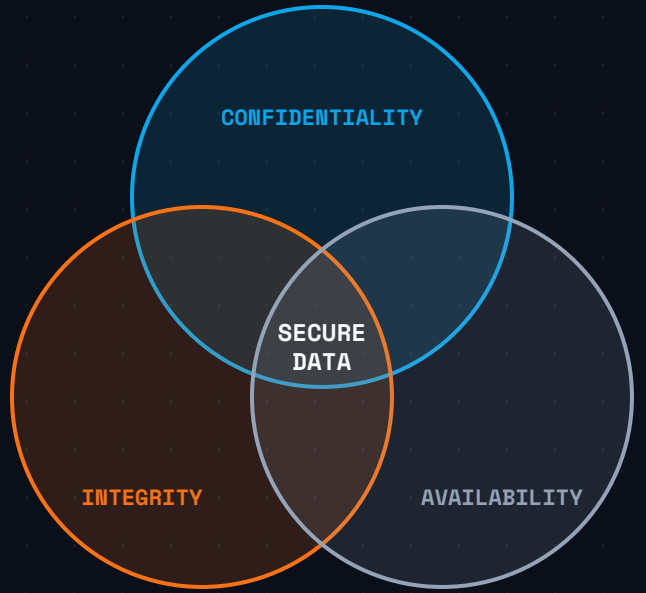
The foundational model for all information security policies.

The CIA Triad is the universally accepted model designed to guide policies for information security. Every security control can be mapped back to these three core principles.

Confidentiality: Data is accessed only by authorized individuals via encryption, strong passwords, and 2FA.

Integrity: Information is trustworthy and accurate; maintained by file permissions and cryptographic hashing.

Availability: Systems are accessible when needed through hardware maintenance, redundancy, and DDoS mitigation.



MODERN CYBER THREATS

Threat actors utilize malicious tools to compromise systems, steal data, and disrupt operations.

MALWARE

Malicious software designed to cause damage or gain unauthorized access. Includes viruses infecting files, self-replicating worms, and trojans disguised as legitimate software.

PHISHING

Fraudulent impersonation via email or text to trick individuals into revealing sensitive data like login credentials or credit card numbers for data theft.

RANSOMWARE

A highly destructive malware that encrypts files or locks systems. Attackers demand ransom payments, typically in cryptocurrency, in exchange for decryption keys.

Social engineering exploits human psychology

The human element remains the weakest link in the security chain.

Despite advanced firewalls and encryption, attackers manipulate emotions—fear, curiosity, or a desire to help—to bypass technical defenses.

Spear-Phishing: Targeted attacks using personalized social media data.

Baiting: Using malware-infected USBs and human curiosity.

Pretexting: Fabricating scenarios to trick victims into access.

INCIDENTS INVOLVING HUMAN ELEMENT

OVER 80%

ATTACK LIFECYCLE





VISUAL EVIDENCE: BAITING

MALWARE-INFECTED MEDIA RELIES ON CURIOSITY TO BYPASS AIR-GAPPED SYSTEMS.

NETWORK SECURITY

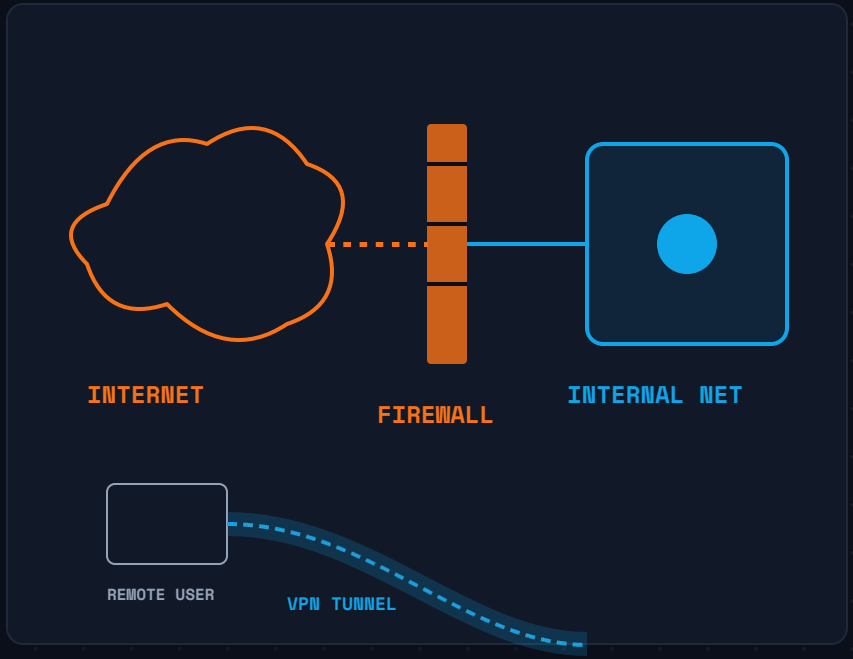
Establishing a secure perimeter and encrypted communication channels.

Network security implements hardware and software mechanisms to protect usability and integrity. It acts as the first line of defense against external threats.

Firewalls: Monitors and filters traffic based on established security policies.

VPNs: Creates secure, encrypted tunnels for remote access over untrusted networks.

Secure Protocols: Uses TLS/SSL (HTTPS) to ensure data in transit remains private.



Cryptography & Data Privacy

Ensuring confidentiality through symmetric and asymmetric encryption models.

Cryptography converts readable **plaintext** into unreadable **ciphertext**, providing the mathematical foundation for modern security.

Data at Rest: Protecting information stored on hard drives and databases.

Data in Transit: Securing data as it moves across public and private networks.

Symmetric Encryption



Uses a **single shared key** for both encryption and decryption. Fast and efficient for large data volumes (e.g., AES), but requires secure key exchange.

Asymmetric Encryption



Uses a **mathematically linked pair**: a Public Key for locking and a Private Key for unlocking. Solves key distribution (e.g., RSA, HTTPS).

The OWASP Top 10

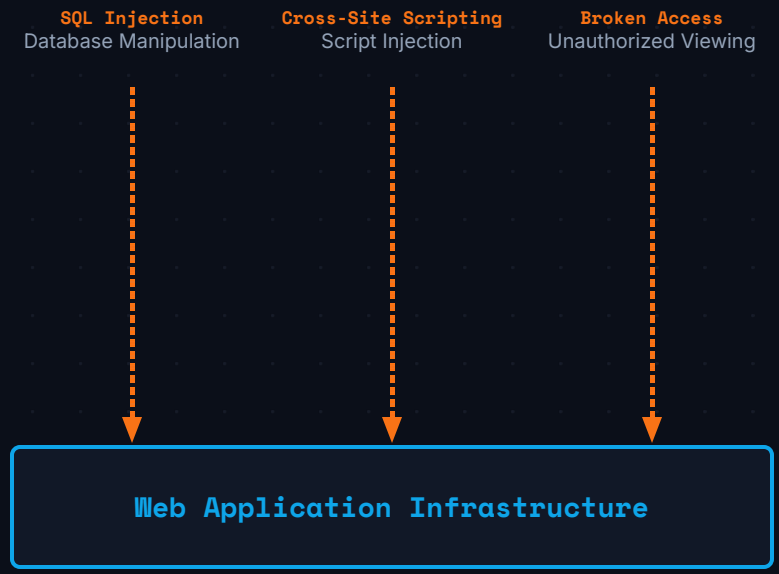
The Open Worldwide Application Security Project (OWASP) maintains a consensus-driven list of the most dangerous flaws.

Understanding these critical vulnerabilities is essential for developing secure software and protecting organizational data assets.

Broken Access Control: Failure to properly enforce user restrictions, leading to unauthorized access.

Injection Flaws: Manipulated database queries (e.g., SQLi) used to steal or destroy backend data.

XSS: Malicious client-side scripts injected into web pages to bypass access controls or steal sessions.



Identity and Access Management

Ensuring the right individuals access the right resources.

Identity and Access Management (IAM) is the framework of policies and technologies ensuring that proper people in an enterprise have the appropriate access to technology resources.

It prevents unauthorized access while facilitating seamless workflows for legitimate users.

System Objective:

Zero Trust Foundation: Verify explicitly, grant least privilege.

Authentication (Who you are)



Verifying a user's identity via MFA: something you **know** (password), something you **have** (token), and something you **are** (biometrics).

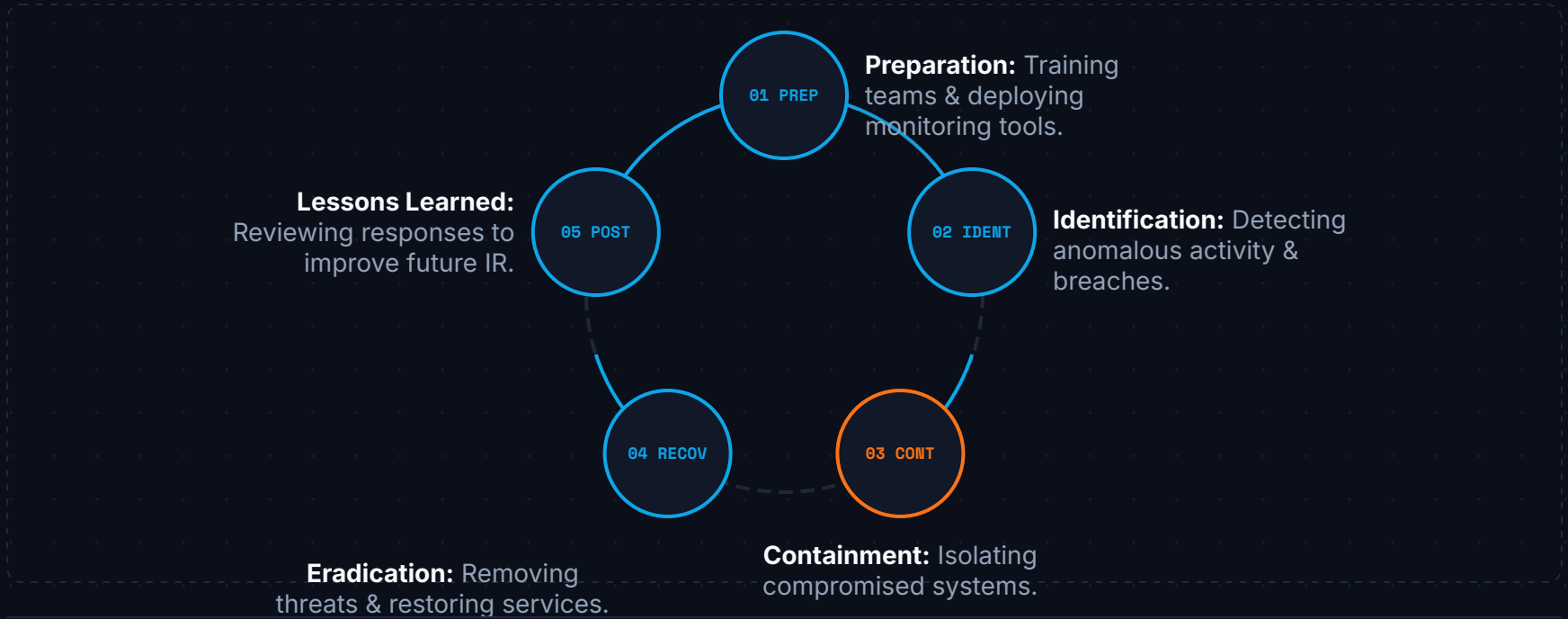
Authorization (What you can do)



Granting specific permissions based on the **Principle of Least Privilege (PoLP)** and **Role-Based Access Control (RBAC)** for data strictly necessary for the job.

A structured incident response plan minimizes damage and accelerates recovery.

Organizations must have a formalized IR plan to detect, manage, and recover from security events efficiently.



CYBERSECURITY EVOLUTION & CAREERS

Rapid field expansion driven by AI and Zero Trust architectures.

FUTURE TRENDS

The future is defined by automation and the transition to **Zero Trust Architectures**—where no user or device is trusted by default.

AI & Machine Learning: Deployed to detect anomalies and respond to threats at machine speed.

Global Talent Gap: Shortage of over **3.4 million** professionals worldwide.

